

SUBJECTIVE QUESTIONS

1. A point moves rectilinearly with a resisting acceleration that depends on the velocity v of the particle as $a = -\alpha v^{1/2}$ where α is a positive constant. At $t = 0$, the velocity is v_0 . What distance will it traverse before it stops? What time will it take to cover that distance?

$$\text{Solution } a = \frac{dv}{dt} = -\alpha v^{1/2} \Rightarrow \int_{v_0}^v \frac{dv}{v^{1/2}} = \int_0^t -\alpha dt$$

$$\Rightarrow 2(\sqrt{v} - \sqrt{v_0}) = -\alpha t$$

$$\text{When } v = 0, \quad t = 2\sqrt{v_0}/\alpha$$

$$\text{Hence, the particle will stop at time } t_0 = 2\sqrt{v_0}/\alpha$$

From above,

$$v = \left(\sqrt{v_0} - \frac{\alpha}{2} t \right)^2 \Rightarrow \int_0^{s_0} ds = \int_0^{t_0} \left(\sqrt{v_0} - \frac{\alpha}{2} t \right)^2 dt$$

$$\therefore s_0 = -\frac{2}{3\alpha} \left(\sqrt{v_0} - \frac{\alpha}{2} t \right)^3 \Big|_0^{t_0}$$

$$= -\frac{2}{3\alpha} \left[\left(\sqrt{v_0} - \frac{\alpha}{2} \times \frac{2\sqrt{v_0}}{\alpha} \right)^3 - (\sqrt{v_0} - 0)^3 \right]$$

$$= \left(\frac{2}{3\alpha} \right) v_0^{3/2}$$

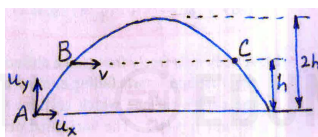
2. A stone is projected from the point on a ground in such a direction so as to hit a bird on the top of a telegraph post of height h and then attain the maximum height $2h$ above the ground. If at the instant of projection, the bird were to fly away horizontally with a uniform speed, find the ratio between the horizontal velocities of the bird and the stone, if the stone still hits the bird while descending.

Solution Let u_x and u_y be respectively the initial horizontal and vertical velocities of the stone and v be the velocity of the bird. If t is the time taken by stone to reach from A to B or C, then

$$h = u_y t - \frac{1}{2} g t^2 \quad \text{and} \quad 2h = \frac{u_y^2}{2g}$$

$$\Rightarrow \frac{u_y^2}{2g} = 2(u_y t - \frac{1}{2} g t^2) \Rightarrow 2g^2 t^2 - 4u_y g t + u_y^2 = 0$$

$$\Rightarrow t = \frac{u_y}{g} \left(\frac{4 \pm \sqrt{16-8}}{4} \right) = \frac{u_y}{2g} (2 \pm \sqrt{2})$$



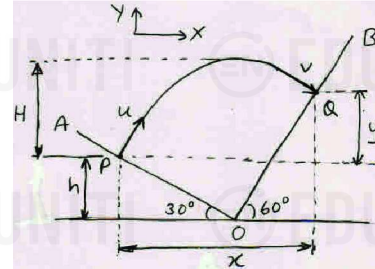
$$\Rightarrow t_{AB} = \frac{u_y}{2g} (2 - \sqrt{2}), \quad t_{AC} = \frac{u_y}{2g} (2 + \sqrt{2})$$

$$t_{BC} = t_{AC} - t_{AB} = \frac{u_y}{2g} (2\sqrt{2})$$

$$\text{Distance } BC = v t_{AC} = u_x t_{BC}$$

$$\therefore \frac{v}{u_x} = \frac{t_{BC}}{t_{AC}} = \frac{2\sqrt{2}}{2 + \sqrt{2}} = \boxed{2(\sqrt{2} - 1)}$$

3. Two inclined planes OA and OB having inclination (with horizontal) 30° and 60° respectively, intersect each other at O as shown in fig.



A particle is projected from point P with velocity

$u = 10\sqrt{3} \text{ ms}^{-1}$ along a direction perpendicular to plane OA. If the

particle strikes plane OB perpendicularly at Q, calculate the

- time of flight.
- speed with which particle strikes the plane OB.
- vertical height h of P from O.
- distance PQ.
- maximum height from O, attained by the particle.

$$\text{Solution } u = 10\sqrt{3} \text{ m s}^{-1}$$

For the path $P \rightarrow Q$,

$$\vec{u} = u(\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j}) = 5\sqrt{3} \hat{i} + 15 \hat{j}$$

$$\vec{v} = \vec{u} + \vec{a} t = (5\sqrt{3} \hat{i} + 15 \hat{j}) + (-10 \hat{j}) t = 5\sqrt{3} \hat{i} + (15 - 10t) \hat{j}$$

(a) Since $\vec{u} \perp \vec{v}$, $\vec{u} \cdot \vec{v} = 0$

$$\Rightarrow (5\sqrt{3} \hat{i} + 15 \hat{j}) \cdot [5\sqrt{3} \hat{i} + (15 - 10t) \hat{j}] = 0$$

$$\Rightarrow 75 + 15(15 - 10t) = 0 \quad \therefore t = 2 \text{ sec}$$

(b) $\vec{v} = 5\sqrt{3} \hat{i} + (15 - 10 \times 2) \hat{j} = 5\sqrt{3} \hat{i} - 5 \hat{j}$

$$\therefore |\vec{v}| = \sqrt{(5\sqrt{3})^2 + 5^2} = \boxed{10 \text{ ms}^{-1}}$$

(c) $x = u_x t = 5\sqrt{3} \times 2 = 10\sqrt{3} \text{ m}$,

$$y = u_y t - \frac{1}{2} g t^2 = 15 \times 2 - \frac{1}{2} \times 10 \times 2^2 = 10 \text{ m}$$

$$OP \cos 30^\circ + OQ \cos 60^\circ = x$$

$$\text{and } OQ \sin 60^\circ - OP \sin 30^\circ = y$$

$$\Rightarrow \frac{\sqrt{3}}{2} OP + \frac{OQ}{2} = 10\sqrt{3} \quad \text{and} \quad \frac{\sqrt{3}}{2} OQ - \frac{OP}{2} = 10$$

$$\text{On solving, we get } OP = 10 \text{ m and } OQ = 10\sqrt{3} \text{ m}$$

$$\therefore h = OP \sin 30^\circ = 10 \times \frac{1}{2} = \boxed{5 \text{ m}}$$

$$(d) \text{ Distance } PQ = \sqrt{x^2 + y^2} = \sqrt{(10\sqrt{3})^2 + 10^2} = \boxed{20 \text{ m}}$$

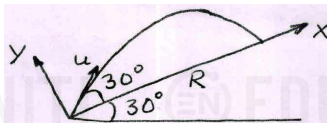
$$(e) H = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{(10\sqrt{3})^2 \times (3/4)}{2 \times 10} = 11.25 \text{ m}$$

Maximum height from O attained by particle is

$$H + h = 11.25 + 5 = \boxed{16.25 \text{ m}}$$

4. A particle is projected with a velocity of 30 m/s at an angle 60° above the horizontal on a slope of inclination 30° . Find its range, time of flight and angle of hit. (Take $g = 10 \text{ m/s}^2$)

Solution Let X and Y axes be as shown in figure.



	u	v	a	s	t
X-axis	$u \cos 30^\circ$	v_x	$-g \sin 30^\circ$	R	T
Y-axis	$u \sin 30^\circ$	v_y	$-g \cos 30^\circ$	0	T

Along Y-axis, $0 = u \sin 30^\circ T - \frac{1}{2} g \cos 30^\circ T^2$

$$\therefore T = \frac{2u}{g} \tan 30^\circ = \frac{2 \times 30}{10} \times \frac{1}{\sqrt{3}} = \boxed{2\sqrt{3} \text{ s}}$$

$$v_y = u \sin 30^\circ - g \cos 30^\circ T = 30 \times \frac{1}{2} - 10 \times \frac{\sqrt{3}}{2} \times 2\sqrt{3} = -15 \text{ m/s}$$

Along X-axis,

$$R = u \cos 30^\circ T - \frac{1}{2} g \sin 30^\circ T^2$$

$$= 30 \times \frac{\sqrt{3}}{2} \times 2\sqrt{3} - \frac{1}{2} \times 10 \times \frac{1}{2} \times (2\sqrt{3})^2$$

$$v_x = u \cos 30^\circ - g \sin 30^\circ T$$

$$= 30 \times \frac{\sqrt{3}}{2} - 10 \times \frac{1}{2} \times 2\sqrt{3} = 5\sqrt{3} \text{ m/s}$$

If the angle of hit is α , then

$$\tan \alpha = |v_y / v_x| = |-15 / 5\sqrt{3}| = \sqrt{3}$$

$$\therefore \boxed{\alpha = 60^\circ}$$

5. Two guns, situated on the top of a tower, fire one shot each with the same speed $5\sqrt{3} \text{ m/s}$ at some interval of time. The first gun fires upward at an angle of 60° with the horizontal and the second gun fires horizontally. The shots collide in air at point P. Find

(a) the coordinates of point P.

(b) the time interval between the firings.

Take the origin of the coordinate system at the point of the firing and trajectories in the x-y plane.

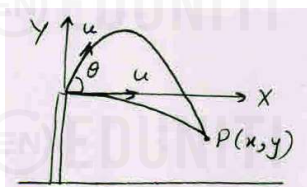
Solution For the first shot,

$$u = 5\sqrt{3} \text{ m/s} \quad \text{and} \quad \theta = 60^\circ$$

$$y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} x^2$$

$$\Rightarrow y = \sqrt{3} x - \frac{2g}{u^2} x^2 \quad \dots (1)$$

For the second shot, $u = 5\sqrt{3} \text{ m/s}$ and $\theta = 0^\circ$



$$\Rightarrow y = -\frac{g}{2u^2} x^2 \quad \dots (2)$$

(a) From equations (1) and (2), we have

$$\sqrt{3} x - \frac{2g}{u^2} x^2 = -\frac{g}{2u^2} x^2 \quad \Rightarrow \sqrt{3} x - \frac{3g}{2u^2} x^2$$

$$\therefore x = \frac{2u^2}{\sqrt{3} g} = \frac{2 \times (5\sqrt{3})^2}{\sqrt{3} \times 10} = 5\sqrt{3} \text{ m}$$

On substituting in eqn (2), we get

$$y = \frac{-10}{2 \times (5\sqrt{3})^2} \times (5\sqrt{3})^2 = -5 \text{ m}$$

The coordinates of point P are $\boxed{(5\sqrt{3} \text{ m}, -5 \text{ m})}$

(b) Time taken by the first and second shots respectively are

$$t_1 = \frac{x}{u \cos \theta} = \frac{5\sqrt{3}}{(5\sqrt{3}) \times 1/2} = 2 \text{ sec} \quad \text{and}$$

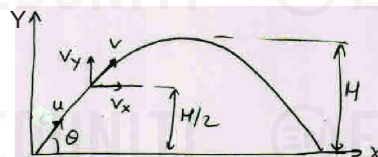
$$t_2 = \frac{x}{u} = \frac{5\sqrt{3}}{(5\sqrt{3})} = 1 \text{ sec}$$

The time interval between the two firings is

$$\boxed{t_1 - t_2 = 2 - 1 = 1 \text{ sec}}$$

6. The speed of the particle when it is at greatest height is A body is $\sqrt{2/5}$ times the speed when it is at half of its greatest height. Find the angle of projection.

Solution



$$H = \frac{u^2 \sin^2 \theta}{2g}$$

At height $H/2$,

$$v_x = u \cos \theta \quad \text{and} \quad v_y^2 = (u \sin \theta)^2 + 2(-g)H/2$$

$$\Rightarrow v_y^2 = u^2 \sin^2 \theta - gH = u^2 \sin^2 \theta - \frac{u^2 \sin^2 \theta}{2} = \frac{u^2 \sin^2 \theta}{2}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{u^2 \cos^2 \theta + \frac{u^2 \sin^2 \theta}{2}}$$

At height H, speed is $v_x = u \cos \theta$

It is given that $v_x = \sqrt{2/5} v$

$$\Rightarrow u \cos \theta = \sqrt{\frac{2}{5}} \times u \sqrt{\cos^2 \theta + \frac{\sin^2 \theta}{2}}$$

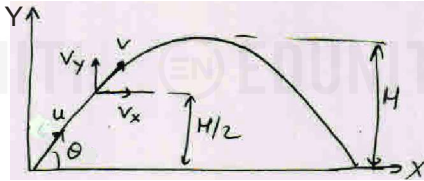
$$\Rightarrow \cos^2 \theta = \frac{2}{5} \left[\cos^2 \theta + \frac{\sin^2 \theta}{2} \right]$$

$$\Rightarrow 5 \cos^2 \theta = 2 \cos^2 \theta + \sin^2 \theta \quad \Rightarrow 3 \cos^2 \theta = \sin^2 \theta$$

$$\Rightarrow \tan^2 \theta = 3$$

$$\therefore \theta = \sin^{-1} \sqrt{2/3}$$

7. A body is thrown at an angle θ with the horizontal such that it attains a speed equal to $\sqrt{2/3}$ times the speed of projection when the body is at half to its maximum height. Find the angle θ
Solution



$$H = \frac{u^2 \sin^2 \theta}{2g}$$

At height $H/2$,

$$v_x = u \cos \theta \quad \text{and} \quad v_y^2 = (u \sin \theta)^2 + 2(-g)H/2$$

$$\Rightarrow v_y^2 = u^2 \sin^2 \theta - gH = u^2 \sin^2 \theta - \frac{u^2 \sin^2 \theta}{2} = \frac{u^2 \sin^2 \theta}{2}$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{u^2 \cos^2 \theta + \frac{u^2 \sin^2 \theta}{2}}$$

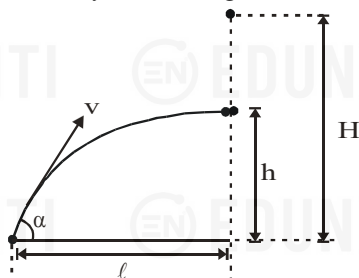
It is given that $v = \sqrt{2/3} u$

$$\Rightarrow \sqrt{\frac{2}{3}} u = u \sqrt{\cos^2 \theta + \frac{\sin^2 \theta}{2}}$$

$$\Rightarrow \frac{2}{3} = \cos^2 \theta + \frac{\sin^2 \theta}{2} = 1 - \frac{\sin^2 \theta}{2}$$

$$\therefore \theta = \sin^{-1} \sqrt{2/3}$$

8. A body falls freely from some altitude H . At the moment the first body starts falling, another body is thrown from the ground which collides with the first at an altitude $h = H/2$. The horizontal distance covered by the second body is l . Find the initial velocity and the angle at which it was thrown.



Solution Let t be the time after which the bodies collide.
For the first body,

$$H - \frac{H}{2} = \frac{1}{2}gt^2 \quad \Rightarrow \quad t = \sqrt{\frac{H}{g}}$$

For the second body,

$$\frac{H}{2} = \frac{v^2 \sin^2 \alpha}{2g} \quad \text{and} \quad l = (v \cos \alpha)t = v \cos \alpha \sqrt{\frac{H}{g}}$$

$$\Rightarrow v \sin \alpha = \sqrt{gH} \quad \text{and} \quad v \cos \alpha = \sqrt{gl^2/H}$$

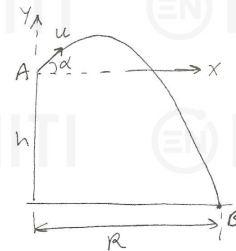
$$\Rightarrow v^2 (\sin^2 \alpha + \cos^2 \alpha) = gH + gl^2/H$$

$$\text{and} \quad \tan \alpha = \frac{\sqrt{gH}}{\sqrt{gl^2/H}} = \frac{H}{l}$$

$$\therefore v = \sqrt{gH \left(1 + \frac{l^2}{H^2} \right)} \quad \text{and} \quad \alpha = \tan^{-1} \frac{H}{l}$$

9. At what angle α above the horizon should a stone be thrown from a steep bank for it to fall into the water as far as possible from the bank? The height of the bank $h = 20$ m and the initial velocity of the stone $u = 15 \text{ ms}^{-1}$. (Take $g = 10 \text{ ms}^{-2}$)

Solution



For the path $A \rightarrow B$, from equation of trajectory, we have

$$-h = (\tan \alpha)R - \frac{g}{2u^2}(1 + \tan^2 \alpha)R^2$$

$$\Rightarrow gR^2 \tan^2 \alpha - 2u^2 R \tan \alpha + (gR^2 - 2hu^2) = 0$$

For $\tan \alpha$ to have real roots, Discriminant ≥ 0

$$\Rightarrow (2Ru^2)^2 - 4gR^2(gR^2 - 2hu^2) \geq 0$$

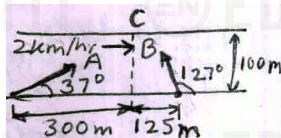
$$\Rightarrow u^4 - g^2 R^2 + 2ghu^2 \geq 0$$

$$\Rightarrow R \leq \frac{u \sqrt{u^2 + 2gh}}{g}$$

$$\therefore R_{\max} = \frac{15 \times \sqrt{15^2 + 2 \times 10 \times 20}}{10} = 37.5 \text{ m}$$

10. Two swimmers start a race. One who reaches point C first on the other bank wins the race. A makes his strokes in a direction of 37° to the river flow with velocity 5 km/hr relative to water. B makes his strokes in a direction 127° to the river flow with same relative velocity. River is flowing with speed of 2 km/hr and is 100 m wide. Who will win the race? Compute the time taken by A and B to reach the point C if the speeds of A and B on the ground are 8 km/hr and 6 km/hr respectively.

Solution



Let t_A and t_B be respectively the times taken by A and B to cross the river. The velocity of A in horizontal direction is

$$5 \cos 37^\circ + 2 = 6 \text{ km/hr}$$

and in vertical direction is

$$5 \sin 37^\circ = 3 \text{ km/hr}$$

The velocity of B in horizontal direction is

$$5 \cos 53^\circ - 2 = 1 \text{ km/hr}$$

and in vertical direction is

$$5 \sin 53^\circ = 4 \text{ km/hr}$$

$$\Rightarrow t_A = \frac{0.1}{3} = \frac{1}{30} \text{ hr} = 120 \text{ s} \quad \text{and}$$

$$t_B = \frac{0.1}{4} = \frac{1}{40} \text{ hr} = 90 \text{ s}$$

The horizontal distances covered by A and B in these times are respectively

$$6 \times 1/30 = 0.2 \text{ km} = 200 \text{ m} \quad \text{and} \quad 1 \times 1/40 = 0.025 \text{ km} = 25 \text{ m}$$

The balance distances that A and B will walk after reaching the other side are respectively

$$s_A = 300 - 200 = 100 \text{ m} \quad \text{and} \quad s_B = 125 - 25 = 100 \text{ m}$$

The time taken by them to walk are respectively

$$t'_A = \frac{0.1}{8} = \frac{1}{80} \text{ hr} = 45 \text{ s} \quad \text{and} \quad t'_B = \frac{0.1}{6} = \frac{1}{60} \text{ hr} = 60 \text{ s}$$

The total time taken to reach point C by A is

$$t_A + t'_A = 120 + 45 = 165 \text{ s}$$

$$\text{and by B is } t_B + t'_B = 90 + 60 = 150 \text{ s}$$

Therefore, B will win the race.

11. During a rainy day, rain is falling vertically with a velocity 2 m s^{-1} . A boy at rest starts his motion with a constant acceleration of 2 m s^{-2} along a straight road. Find the rate at which the angle of the axis of umbrella with vertical should be changed so that the rain always falls parallel to the axis of the umbrella.

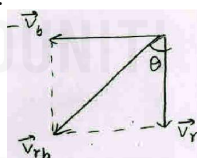
Solution

$$\vec{v}_{rb} = \vec{v}_r - \vec{v}_b \quad \text{and} \quad \tan \theta = \frac{v_b}{v_r}$$

where, $v_r = 2 \text{ m s}^{-1}$ and $v_b = 2t \text{ m s}^{-1}$

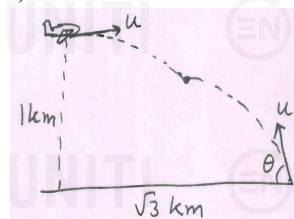
$$\Rightarrow \theta = \tan^{-1} \left(\frac{v_b}{v_r} \right) = \tan^{-1} \left(\frac{2t}{2} \right) = \tan^{-1} t$$

$$\therefore \frac{d\theta}{dt} = \frac{1}{1+t^2}$$



12. An aircraft is flying horizontally with a constant velocity $u = 200 \text{ m/s}$, at a height $= 1 \text{ km}$ above the ground. At the moment shown, a bomb is released from the aircraft and the cannon-gun below fires a shell with initial speed of 200 m/s , at some angle θ .

- (a) For what value of ' θ ' will the projectile shell destroy the bomb in mid-air?
(b) If the value of θ is 53° , find the minimum distance between the bomb and the shell as they fly past each other.
(Take $\sin 53^\circ = 4/5$.)



Solution Here, $u = 200 \text{ m/s}$.

The acceleration of shell w.r.t. to the bomb is zero. Its horizontal and vertical velocities with respect to bomb are respectively

$$v_x = u(1 + \cos \theta) \quad \text{and} \quad v_y = u \sin \theta$$

- (a) If shell destroys the bomb at time t , then

$$t = \frac{\sqrt{3} \text{ km}}{u(1 + \cos \theta)} = \frac{1 \text{ km}}{u \sin \theta}$$

$$\Rightarrow 3 \sin^2 \theta = (1 + \cos \theta)^2$$

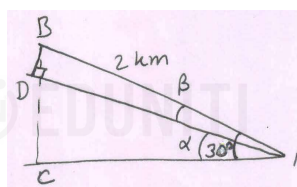
$$\Rightarrow 3(1 + \cos \theta)(1 - \cos \theta) = (1 + \cos \theta)^2$$

$$\Rightarrow 3(1 - \cos \theta) = 1 + \cos \theta$$

$$\Rightarrow \cos \theta = 1/2 \quad \therefore \theta = 60^\circ$$

- (b) If $\theta = 53^\circ$, then $v_x = 200(1 + 0.6) = 320 \text{ m/s}$

$$\text{and} \quad v_y = 200 \times 0.8 = 160 \text{ m/s}$$



$$\Rightarrow \tan \alpha = \frac{v_y}{v_x} = \frac{1}{2}$$

$$\Rightarrow \cos \alpha = \frac{2}{\sqrt{5}} \quad \text{and} \quad \sin \alpha = \frac{1}{\sqrt{5}}$$

The motion of shell w.r.t. the bomb is along AD.

The minimum distance between the bomb and the shell is

$$BD = AB \sin \beta = 2 \sin (30^\circ - \alpha)$$

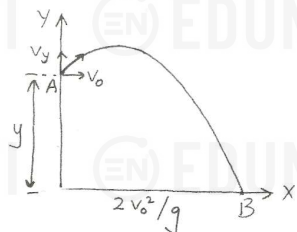
$$= 2[\sin 30^\circ \cos \alpha - \cos 30^\circ \sin \alpha]$$

$$= 2 \left(\frac{1}{2} \times \frac{2}{\sqrt{5}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{5}} \right) = \frac{2 - \sqrt{3}}{\sqrt{5}} \text{ km}$$

$$\approx 120 \text{ m}$$

13. A balloon moves up vertically such that if a stone is thrown from it with a horizontal velocity v_0 relative to it, the stone always hits the ground at a fixed point $2v_0^2/g$ horizontally away from it. Find the height of the balloon as a function of time.

Solution



Consider path $A \rightarrow B$. When the balloon is at height y , the horizontal and vertical velocities of the stone at point A are respectively

$$v_x = v_0 \quad \text{and} \quad v_y = dy/dt$$

Since the stone falls at a horizontal distance $2v_0^2/g$, the time taken by it during the path is

$$t = \frac{2v_0^2/g}{v_0} = \frac{2v_0}{g}$$

Along Y-axis, $-y = v_y t - \frac{1}{2}gt^2$

$$\Rightarrow -y = \frac{dy}{dt} \left(\frac{2v_0}{g} \right) - \frac{g}{2} \left(\frac{2v_0}{g} \right)^2 = \frac{2v_0}{g} \left(\frac{dy}{dt} - v_0 \right)$$

$$\Rightarrow \frac{dy}{dt} = v_0 - \frac{gy}{2v_0} \quad \Rightarrow \int_0^y \frac{dy}{2v_0^2 - gy} = \int_0^t \frac{1}{2v_0} dt$$

$$\Rightarrow -\frac{1}{g} \ln(2v_0^2 - gy) \Big|_0^y = \frac{t}{2v_0}$$

$$\Rightarrow \ln \left(\frac{2v_0^2 - gy}{2v_0^2} \right) = -\frac{gt}{2v_0} \quad \Rightarrow 1 - \frac{gy}{2v_0^2} = e^{-gt/2v_0}$$

$$\therefore y = \frac{2v_0^2}{g} (1 - e^{-gt/2v_0})$$

PARAGRAPH TYPE

Paragraph for Question 14 to 16

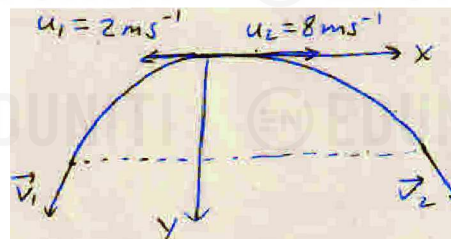
From a tower of height 40 m, two bodies are simultaneously projected horizontally in opposite directions, with velocities 2 m s^{-1} and 8 m s^{-1} , respectively. ($g = 10 \text{ m s}^{-2}$)

14. The time taken for the velocity vectors of two bodies to become perpendicular to each other, is
 (a) 0.1 s (b) 0.2 s
 (c) 0.4 s (d) 0.8 s
15. The horizontal distance between two bodies, when their velocity are perpendicular to each other, is
 (a) 1 m (b) 0.5 m
 (c) 2 m (d) 4 m

16. The time taken for the displacement vectors of two bodies to become perpendicular to each other is
 (a) 0.1 s (b) 0.2 s
 (c) 0.8 s (d) 0.6 s

Solution

14.(c)



Let X and Y axes be as shown in fig.

$$\vec{v}_1 = \vec{u}_1 + \vec{a} t = -2\hat{i} + 10\hat{j} t, \quad \vec{v}_2 = \vec{u}_2 + \vec{a} t = 8\hat{i} + 10\hat{j} t$$

$$\vec{v}_1 \perp \vec{v}_2 \quad \Rightarrow \vec{v}_1 \cdot \vec{v}_2 = 0$$

$$\Rightarrow -16 + 100 t^2 = 0 \quad \therefore t = 0.4 \text{ s.}$$

- 15.(d) Horizontal distance between the two bodies is

$$(u_1 + u_2) t = (2 + 8) \times 0.4 = 4 \text{ m}$$

$$16.(c) \quad \vec{s}_1 = \vec{u}_1 t + \frac{1}{2} \vec{a} t^2 = -2\hat{i} t + \frac{1}{2} \times 10\hat{j} t^2$$

$$\vec{s}_2 = \vec{u}_2 t + \frac{1}{2} \vec{a} t^2 = 8\hat{i} t + \frac{1}{2} \times 10\hat{j} t^2$$

$$\vec{s}_1 \perp \vec{s}_2 \quad \Rightarrow \vec{s}_1 \cdot \vec{s}_2 = 0$$

$$\Rightarrow -16 t^2 + 25 t^4 = 0 \quad \therefore t = 0.8 \text{ s.}$$

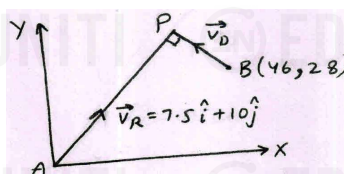
Paragraph for Question 17 to 18

A rabbit is hidden in grass at point A near a pond. A dog is drinking water in the pond at point B. The coordinates of points A and B are (0, 0) and (46 m, 28 m), respectively. At the same instant, rabbit starts to run with a constant velocity $(7.5\hat{i} + 10\hat{j}) \text{ ms}^{-1}$ and dog starts to run with constant speed in a straight line and catches rabbit in time t .

17. If x and y -components of velocities of dog for catching the rabbit are v_1 and v_2 , respectively, which of following relations is correct?
 (a) $7.5t - 46 = v_1 t$ (b) $10t - 28 = v_2 t$
 (c) Both (a) and (b) (d) Neither (a) nor (b)
18. If it runs with minimum speed, the minimum time in which the dog could catch the rabbit is
 (a) 4 s (b) 5.25 s
 (c) 5 s (d) 4.6 s

Solution

17. (c) Velocity of rabbit is $\vec{v}_R = (7.5\hat{i} + 10\hat{j}) \text{ ms}^{-1}$



Velocity of dog is $\vec{v}_D = v_1 \hat{i} + v_2 \hat{j}$

If the dog catches rabbit in time t , then

$$46 = (7.5 - v_1)t \text{ and } 28 = (10 - v_2)t$$

$$\therefore 7.5t - 46 = v_1 t \text{ and } 10t - 28 = v_2 t$$

18. (a) If the dog catches rabbit by running with minimum speed, then the dog must run along BP such that

$$AP \perp BP$$

$$\Rightarrow \text{Slope of AP} \times \text{slope of BP} = -1$$

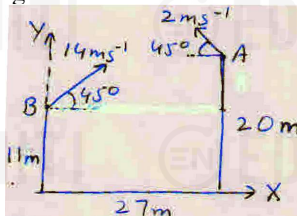
$$\Rightarrow \frac{10}{7.5} \times \frac{v_2}{v_1} = -1$$

$$\Rightarrow 3v_1 + 4v_2 = 0$$

$$\Rightarrow 3(7.5t - 46) + 4(10t - 28) = 0 \quad \therefore t = 4s$$

Paragraph for Question 19 to 21

Two particles are thrown simultaneously from points A and B with velocities $u_1 = 2 \text{ m s}^{-1}$ and $u_2 = 14 \text{ m s}^{-1}$, respectively, as shown in fig.



19. The relative velocity of B as seen from A is

- (a) $8\sqrt{2}\hat{i} + 6\sqrt{2}\hat{j}$ (b) $4\sqrt{2}\hat{i} + 3\sqrt{3}\hat{j}$
(c) $3\sqrt{5}\hat{i} + 2\sqrt{3}\hat{j}$ (d) $3\sqrt{2}\hat{i} + 4\sqrt{3}\hat{j}$

20. Minimum separation between A and B is

- (a) 3 m (b) 6 m
(c) 9 m (d) 12 m

21. Time after which the separation between A and B is minimum is

- (a) 2.7 s (b) $1.35\sqrt{2} \text{ s}$
(c) 2 s (d) 3 s

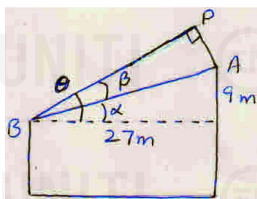
Solution

19. (a)

$$\vec{v}_{BA} = (14 \cos 45^\circ \hat{i} + 14 \sin 45^\circ \hat{j}) - (-2 \cos 45^\circ \hat{i} + 2 \sin 45^\circ \hat{j})$$

$$= 8\sqrt{2}\hat{i} + 6\sqrt{2}\hat{j}$$

20. (c)



$$AB = 8\sqrt{27^2 + 9^2} = 9\sqrt{10} \text{ m} \quad \tan \theta = 6\sqrt{2}/8\sqrt{2} = 3/4$$

$$\tan \alpha = 9/27 = 1/3$$

$$\Rightarrow \tan \beta = \tan (\theta - \alpha) = \frac{3/4 - 1/3}{1 + 3/4 \times 1/3} = \frac{1}{3}$$

$$\Rightarrow \sin \beta = 1/\sqrt{10} \text{ and } \cos \beta = 3/\sqrt{10}$$

Minimum separation

$$= AP = AB \sin \beta = 9\sqrt{10} \times 1/\sqrt{10} = 9 \text{ m}$$

$$21. (b) BP = AB \cos \beta = 9\sqrt{10} \times 3/\sqrt{10} = 27 \text{ m}$$

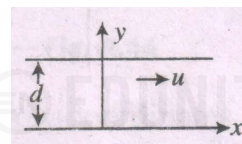
$$|\vec{v}_{BA}| = \sqrt{(8\sqrt{2})^2 + (6\sqrt{2})^2} = 10\sqrt{2} \text{ ms}^{-1}$$

Time after which separation is minimum is

$$t = \frac{BP}{|\vec{v}_{BA}|} = \frac{27}{10\sqrt{2}} = 1.35\sqrt{2} \text{ s}$$

Paragraph for Question 22 to 24

A river of width d is flowing with velocity u . A boat starts from rest and with respect to water, moves perpendicular to the river with an acceleration of $a = 5t$, where t is time. The boat starts from point $(1, 0)$ of the coordinate system as shown in fig.



22. Find the equation of trajectory of the boat.

- (a) $y = \frac{6}{5} \left(\frac{x}{u} \right)^3$ (b) $y = \frac{5}{6} \left(\frac{x}{u} \right)^3$
(c) $y = \frac{5}{6} = u \left(\frac{x-1}{u} \right)^3$ (d) $y = \frac{6}{5} \left(\frac{x-1}{u} \right)^3$

23. Find the drift of the boat when it is in the middle of the river.

- (a) $u \left(\frac{3d}{5} \right)^{1/3}$ (b) $u \left(\frac{3d}{5} \right)^{1/3} + 1$
(c) $u \left(\frac{5d}{6} \right)^{1/3}$ (d) $u \left(\frac{5d}{6} \right)^{1/3} + 1$

24. Obtain the total time taken to cross the river.

- (a) $(3d/5)^{1/3}$ (b) $(6d/5)^{1/3}$
(c) $(5d/3)^{1/3}$ (d) $(5d/6)^{1/3}$

Solution

$$22. (c) a_y = 5t$$

$$\Rightarrow \int_0^{v_y} dv_y = \int_0^t 5t \, dt$$

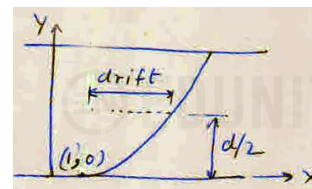
$$\Rightarrow v_y = \frac{dy}{dt} = \frac{5t^2}{2}$$

$$\Rightarrow \int_0^y dy = \int_0^t \frac{5t^2}{2} dt$$

$$\Rightarrow y = \frac{5}{6} t^3$$

$$\text{Also, } x = 1 + ut$$

$$\therefore y = \frac{5}{6} \left(\frac{x-1}{u} \right)^3$$





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$$23(a) \quad (x-1) = u \left(\frac{6y}{5} \right)^{1/3}$$

When the boat is in the middle of river, $y = d/2$

$$\therefore \text{Drift} = x-1 = u \left(\frac{6 \times d/2}{5} \right)^{1/3} = u \left(\frac{3d}{5} \right)^{1/3}$$

$$24(b) \quad y = \frac{5}{6} t^3 \quad \Rightarrow \quad t = \left(\frac{6y}{5} \right)^{1/3}$$

When the boat crosses the river, $y = d$.

Therefore, time taken is $(6d/5)^{1/3}$

Paragraph for Question 25 to 27

A projectile is projected with some initial velocity and some initial angle of projection. Wind is also blowing, due to which constant horizontal retardation a is imparted to the particle in the plane of motion. It is found that the particle is at same height at two different time t_1 and t_2 and particle is at same horizontal distance at two different time t_3 and t_4

25. Angle of projection of particle is

$$(a) \tan^{-1} \left(\frac{t_1 + t_2}{t_3 + t_4} \right) \quad (b) \tan^{-1} \left[\frac{a(t_1 + t_2)}{g(t_3 + t_4)} \right]$$

$$(c) \tan^{-1} \left[\frac{g(t_1 + t_2)}{a(t_3 + t_4)} \right] \quad (d) \text{None of these}$$

26. Maximum height of the particle is

$$(a) \frac{g}{8} (t_1 + t_2)^2 \quad (b) \frac{g}{4} (t_1 + t_2)^2$$

$$(c) \frac{g}{2} (t_1 + t_2)^2 \quad (d) \text{None of these}$$

27. Range of the projectile is

$$(a) \frac{1}{2} a(t_1 + t_2)^2 \quad (b) \frac{1}{2} a(t_1 + t_2)(t_3 + t_4 - t_1 - t_2)$$

$$(c) \frac{1}{2} a(t_3 + t_4 - t_1 - t_2)^2$$

$$(d) \text{can't be determined}$$

Solution

25 (c) Let u be the initial velocity and θ be the angle of projection.

Then,

$$x = u \cos \theta t - \frac{1}{2} a t^2 \quad \text{and} \quad y = u \sin \theta t - \frac{1}{2} g t^2$$

$$\Rightarrow at^2 - 2u \cos \theta t + 2x = 0 \quad \text{and}$$

$$\Rightarrow gt^2 - 2u \sin \theta t + 2y = 0$$

Let the particle be at horizontal distance x at time t_3 and t_4 and at height y at time t_1 and t_2 .

Then the sum of roots of the equations are

$$t_3 + t_4 = \frac{2u \cos \theta}{a} \quad \text{and} \quad t_1 + t_2 = \frac{2u \sin \theta}{g}$$

$$\Rightarrow \frac{t_1 + t_2}{t_3 + t_4} = \frac{a \tan \theta}{g} \quad \therefore \theta = \tan^{-1} \left[\frac{g(t_1 + t_2)}{a(t_3 + t_4)} \right]$$

26 (a) At maximum height, $v_y = 0$

$$\Rightarrow 0^2 - (u \sin \theta)^2 = 2(-g)h_{\max}$$

$$\therefore h_{\max} = \frac{(u \sin \theta)^2}{2g} = \frac{[g(t_1 + t_2)/2]^2}{2g} = \frac{g}{8} (t_1 + t_2)^2$$

$$27 (b) \text{ Time of flight, } T = \frac{2u \sin \theta}{g} = t_1 + t_2$$

$$\begin{aligned} \text{Range, } R &= (u \cos \theta)T - \frac{1}{2} a T^2 \\ &= \frac{a}{2} (t_3 + t_4)(t_1 + t_2) - \frac{a}{2} (t_1 + t_2)^2 \\ &= \frac{a}{2} (t_1 + t_2)(t_3 + t_4 - t_1 - t_2) \end{aligned}$$